

Exponential distribution

Defn:-

A continuous random variable X is said to follow an exponential distribution with parameter $\lambda > 0$, if its probability density function is given by

$$f(x) = \begin{cases} \lambda e^{-\lambda x} & x \geq 0 \\ 0 & \text{otherwise} \end{cases}$$

r^{th} moment of exponential distribution

$$E(X^r) = \int_0^{\infty} x^r \lambda e^{-\lambda x} dx = \lambda \int_0^{\infty} e^{-\lambda x} x^r dx$$

$$\text{put } y = \lambda x \text{ then } dy = \lambda dx \Rightarrow dx = \frac{dy}{\lambda}$$

$$\therefore E(X^r) = \lambda \int_0^{\infty} e^{-y} \left(\frac{y}{\lambda}\right)^r \frac{dy}{\lambda}$$

$$= \frac{1}{\lambda^r} \int_0^{\infty} e^{-y} y^r dy = \frac{1}{\lambda^r} \int_0^{\infty} e^{-y} y^{(r+1)-1} dy$$

$$[\text{Recall:- } \Gamma(n) = \int_0^{\infty} e^{-x} x^{n-1} dx]$$

$$\therefore E(X^r) = \frac{1}{\lambda^r} \Gamma(r+1)$$

Hence

$$E(X^r) = \frac{r!}{\lambda^r}$$

Mean $E(X) = \frac{1!}{\lambda^1} = \frac{1}{\lambda}$

$$E(X^2) = \frac{2!}{\lambda^2} = \frac{2}{\lambda^2}$$

$$\text{Var}(X) = E(X^2) - [E(X)]^2 = \frac{2}{\lambda^2} - \frac{1}{\lambda^2} = \frac{1}{\lambda^2}$$

Memory less property of exponential distribution

Let X be a random variable which is exponentially distributed, then

$$P(X > s+t | X > s) = P(X > t) \text{ for any } s, t > 0$$

Proof:-

$$P(X > s+t | X > s) = \frac{P((X > s+t) \cap (X > s))}{P(X > s)}$$



$$\text{Thus } P(X > s+t | X > s) = \frac{P(X > s+t)}{P(X > s)}$$

$$= \frac{\int_{s+t}^{\infty} \lambda e^{-\lambda x} dx}{\int_s^{\infty} \lambda e^{-\lambda x} dx} = \frac{\left[\frac{e^{-\lambda x}}{-\lambda} \right]_{s+t}^{\infty}}{\left[\frac{e^{-\lambda x}}{-\lambda} \right]_s^{\infty}}$$

$$= \frac{e^{-\lambda(s+t)}}{e^{-\lambda s}} = e^{-\lambda t} = P(X > t)$$

Pb:1 The mileage which car owner get with a certain kind of radial tire is a RV having an exponential distribution with mean 40,000 km. Find the probabilities that one of these tires will last (i) at least 20,000 km and (ii) at most 30,000 km.

Soln:- Let X denote the mileage obtained with the tire.

Given mean = 40,000

$$(a) \frac{1}{\lambda} = 40,000 \Rightarrow \lambda = \frac{1}{40,000}$$

$$\text{Now, } f(x) = \lambda e^{-\lambda x} \quad x > 0$$

$$\therefore f(x) = \frac{1}{40,000} e^{-\frac{x}{40,000}} \quad x > 0$$

$$\begin{aligned} (i) P(X \geq 20,000) &= \int_{20,000}^{\infty} f(x) dx \\ &= \int_{20,000}^{\infty} \frac{1}{40,000} e^{-\frac{x}{40,000}} dx \\ &= \frac{1}{40,000} \left[\frac{e^{-\frac{x}{40,000}}}{\frac{-1}{40,000}} \right]_{20,000}^{\infty} \\ &= e^{-1/2} = 0.6065 \end{aligned}$$

$$\begin{aligned}
 \text{(ii) } P(X \leq 30,000) &= \int_0^{30,000} \frac{1}{40,000} e^{-\frac{x}{40,000}} dx \\
 &= \frac{1}{40,000} \left[\frac{e^{-\frac{x}{40,000}}}{-\frac{1}{40,000}} \right]_0^{30,000} \\
 &= 1 - e^{-3/4} = 0.5276.
 \end{aligned}$$

Pb: 2 The time (in hours) required to repair a machine is exponentially distributed with parameter $\lambda = 1/2$.

(a) What is the probability that the repair time exceeds 2 h?

(b) What is the conditional probability that a repair takes at least 10 h given that its duration exceeds 9 h?

Soln:- $f(x) = \lambda e^{-\lambda x}, x > 0$

$$= \frac{1}{2} e^{-\frac{x}{2}} \quad x > 0$$

$$(i) P(X > 2) = \int_2^{\infty} f(x) dx = \int_2^{\infty} \frac{1}{2} e^{-x/2} dx$$
$$= \left[e^{-x/2} \right]_2^{\infty} = e^{-1} = 0.3678.$$

(b) ^{Find} $P(X > 10 | X > 9)$. By the memory less property of exponential distribution, we have

$$\begin{aligned} P(X > 10 | X > 9) &= P(X > 1) \\ &= \int_1^{\infty} \frac{1}{2} e^{-x/2} dx = \left[-e^{-x/2} \right]_1^{\infty} \\ &= e^{-1/2} = 0.6065. \end{aligned}$$